

EXERCISE 12

Intro to Constraints: NOT NULL and UNIQUE Constraints

Global Fast Foods has been very successful this past year and has opened several new stores. They need to add a table to their database to store information about each of their store's locations. The owners want to make sure that all entries have an identification number, date opened, address, and city and that no other entry in the table can have the same email address. Based on this information, answer the following questions about the global_locations table. Use the table for your answers.

NAME	TYPE	LENGTH	PRECISION	SCALE	NULLABLE	DEFAULT
Id						
name						
date_opened						
address						
city						
zip/postal code						
phone						
email						
manager_id						
Emergency contact						

1. What is a "constraint" as it relates to data integrity?

A constraint is a rule that ensures data accuracy and integrity in a table.

2. What are the limitations of constraints that may be applied at the column level and at the table level?

Column level constraints apply to one column; table-level can apply to many.

3. Why is it important to give meaningful names to constraints?

Meaningful names make constraints easy to identify and manage.

4. Based on the information provided by the owners, choose a datatype for each column. Indicate the length, precision, and scale for each NUMBER datatype.

Id - NUMBER (6) NOT NULL | address - VARCHAR2 (100) NOT NULL
 name - VARCHAR (50) NOT NULL | city - VARCHAR (30) NOT NULL
 date_opened - DATE NOT NULL | phone - VARCHAR (15) (nullable)
 zip - postal - code - VARCHAR (10) (nullable)

5. Use "(nullable)" to indicate those columns that can have null values.

Nullable Columns:

zip - postal - code, phone, manager_id, emergency - contact.

6. Write the CREATE TABLE statement for the Global Fast Foods locations table to define the constraints at the column level.

CREATE TABLE Global-locations(
 id NUMBER(4) PRIMARY KEY, loc_name VARCHAR(20) NOT NULL,
 date-opened DATE NOT NULL, address VARCHAR2(30) NOT NULL,
 city VARCHAR2(20) NOT NULL, zip_postal VARCHAR2(20), phone VARCHAR2(15),
 email VARCHAR2(40) UNIQUE, contact VARCHAR2(40));

7. Execute the CREATE TABLE statement in Oracle Application Express.

execute this sql in oracle application express window.

8. Execute a DESCRIBE command to view the Table Summary information.

DESC global-locations;

9. Rewrite the CREATE TABLE statement for the Global Fast Foods locations table to define the UNIQUE constraints at the table level. Do not execute this statement.

NAME	TYPE	LENGTH	PRECISION	SCALE	NULLABLE	DEFAULT
id	number	4				
loc_name	varchar2	20			X	
	date					
address	varchar2	30				
city	varchar2	20				
zip_postal	varchar2	20			X	
phone	varchar2	15			X	
email	varchar2	40			X	
manager_id	number	4			X	
contact	varchar2	40			X	

CREATE TABLE global-locations C

id NUMBER(4),
 loc_name VARCHAR2(20) NOT NULL,
 date-opened DATE NOT NULL,
 address VARCHAR2(30) NOT NULL,
 city VARCHAR2(20) NOT NULL,
 zip_postal VARCHAR2(20), phone VARCHAR2(15), manager_id NUMBER,
 contact VARCHAR2(40)
 Constraint pk-global-id Primary key (id),
 Constraint uq-global-email Unique (email));

PRIMARY KEY, FOREIGN KEY, and CHECK Constraints

- What is the purpose of a
 - PRIMARY KEY - Uniquely identifies each row in a table
 - FOREIGN KEY - Links one table to another using a related column
 - CHECK CONSTRAINT - ensures values meet specific conditions

- Using the column information for the animals table below, name constraints where applicable at the table level, otherwise name them at the column level. Define the primary key (animal_id). The license_tag_number must be unique. The admit_date and vaccination_date columns cannot contain null values.

animal_id NUMBER(6)
 name VARCHAR2(25)
 license_tag_number NUMBER(10)
 admit_date DATE
 adoption_id NUMBER(5),
 vaccination_date DATE

animal_id → Primary Key

license_tag_number → Unique

admit_date

vaccination_date → NOT NULL

- Create the animals table. Write the syntax you will use to create the table.

CREATE TABLE animals (

animal_id number (6) Primary key,

name Varchar2(25), license_tag_number Number(10) unique,
 admit_date Date NOT NULL, adoption_id NUMBER(5),
 vaccination_date DATE NOT NULL);

- Enter one row into the table. Execute a SELECT statement to verify your input. Refer to the graphic below for input.

ANIMAL_ID	NAME	LICENSE_TAG_NUMBER	ADMIT_DATE	ADOPTION_ID	VACCINATION_DATE
101	Spot	35540	10-Oct-2004	205	12-Oct-2004

INSERT INTO animals VALUES (101, 'Spot', 35540,
 '10-Oct-2004', 205, '12-Oct-2004')

SELECT * FROM animals;

- Write the syntax to create a foreign key (adoption_id) in the animals table that has a corresponding primary-key reference in the adoptions table. Show both the column-level and table-level syntax. Note that because you have not actually created an adoptions table, no adoption_id primary key exists, so the foreign key cannot be added to the animals table.

adoption_id NUMBER(5) REFERENCES adoptions

(adoption_id)

Constraint yfk-adopt Foreign Key (adoption_id)

REFERENCES adoptions (adoption_id)

6. What is the effect of setting the foreign key in the ANIMAL table as:

- a. ON DELETE CASCADE
- b. ON DELETE SET NULL

ON Delete (cascade-) when the referenced (Parent) record is deleted all related records in the Animal table are also deleted automatically

ON DELETE SET NULL → when the referenced (Parent) record is deleted the foreign key field in the Animal is set to NULL

7. What are the restrictions on defining a CHECK constraint?

The Condition must be a Boolean expression

It cannot reference other tables or subqueries

It applies only to the current row's columns

It cannot use aggregated functions (like SUM, AVG)

Evaluation Procedure	Marks awarded
Query(5)	5
Execution (5)	5
Viva(5)	5
Total (15)	15
Faculty Signature	R ₇ h